Unit I :: Frames of Reference

PHY01MX01R04: Mechanics and Properties of Matter

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Syllabus

Unit I: Frames of Reference

Review of Newton's laws of motion; inertial frames of reference, Galilean transformations, Galilean invariance; non-inertial frames of reference, fictitious (inertial) forces – centrifugal force, Coriolis force; transformation equations in inclined and rotating frames.

1 Review of Newton's laws of motion

Before delving into a review of Newton's laws of motion, we need to be clear of the definitions of some important terms used in mechanics:

- *Particle:* an idealized object that has mass and position but occupies zero volume.
- *Event:* a specific occurrence that happens at a unique place in space and a specific instant in time.
- *Observer:* a person or an equipment which can locate, record, measure and interpret an event.
- *Frame of reference:* a coordinate system combined with a clock, used by an observer to measure the position and motion of particles.
- *Rest:* a particle is said to be at rest when its position does not change relative to a chosen frame of reference over time.
- *Motion:* the change in position or orientation of an object with respect to a chosen frame of reference over time.
- *Inertia:* the tendency of a body to resist any change in its motion or configuration.
- *Position:* the location of a particle in space with respect to a fixed reference point known as the origin. If (x, y, z) are the coordinates of a particle in Cartesian space, then its position vector is

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}} \tag{1}$$

- *Displacement:* the change in the position of a particle. If a particle is initially at the position $\mathbf{r}_i = x_i\hat{\mathbf{i}} + y_i\hat{\mathbf{j}} + z_i\hat{\mathbf{k}}$ and then finally moves to the position $\mathbf{r}_f = x_f\hat{\mathbf{i}} + y_f\hat{\mathbf{j}} + z_f\hat{\mathbf{k}}$, its displacement vector is

$$\Delta\mathbf{r} = \mathbf{r}_f - \mathbf{r}_i = (x_f - x_i)\hat{\mathbf{i}} + (y_f - y_i)\hat{\mathbf{j}} + (z_f - z_i)\hat{\mathbf{k}} \tag{2}$$

  If the displacement is infinitessimally small, then we can write

$$\mathrm{d}\mathbf{r} = \hat{\mathbf{i}}\,\mathrm{d}x + \hat{\mathbf{j}}\,\mathrm{d}y + \hat{\mathbf{k}}\,\mathrm{d}z \tag{3}$$

- *Velocity:* the rate of change of position of a particle. If the position of a particle varies with time, then its instantaneous velocity is the first derivative of its position with respect to time.

$$\mathbf{v} = \dot{\mathbf{r}} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \tag{4}$$

  If the positions of a particle at two instants of time are $\mathbf{r}(t_1) = \mathbf{r}_1$ and $\mathbf{r}(t_2) = \mathbf{r}_2$, then its displacement is obtained by the integral

$$\begin{aligned} \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathrm{d}\mathbf{r} &= \int_{t_1}^{t_2} \mathbf{v}\,\mathrm{d}t \\ \implies \mathbf{r}_2 - \mathbf{r}_1 &= \int_{t_1}^{t_2} \mathbf{v}\,\mathrm{d}t \end{aligned} \tag{5}$$

- *Acceleration:* the rate of change of velocity of a particle. If the velocity of a particle varies with time, then its instantaneous acceleration is the first derivative of its velocity with respect to time.

$$\mathbf{a} = \dot{\mathbf{v}} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \tag{6}$$

$$\text{or} \quad \mathbf{a} = \ddot{\mathbf{r}} = \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} \tag{7}$$

  If the velocities of a particle at two instants of time are $\mathbf{v}(t_1) = \mathbf{v}_1$ and $\mathbf{v}(t_2) = \mathbf{v}_2$, then the change in its velocity is obtained by the integral

$$\int_{\mathbf{v}_1}^{\mathbf{v}_2} \mathrm{d}\mathbf{v} = \int_{t_1}^{t_2} \mathbf{a}\,\mathrm{d}t$$

$$\implies \mathbf{v}_2 - \mathbf{v}_1 = \int_{t_1}^{t_2} \mathbf{a}\,\mathrm{d}t \tag{8}$$

- *Momentum:* the product of the mass and velocity of a body, i.e. $\mathbf{p} = m\mathbf{v}$. In mechanics, we also refer to the momentum of a body as its *amount of motion*.

1.1 Kinematics relations

The kinematic relations are a set of formulae in mechanics that describe the motion of objects moving with constant acceleration, without considering the forces that cause the motion.

1.1.1 Velocity-time relation

Consider the motion of a particle moving with a constant acceleration $\mathbf{a}$. Let its initial velocity be $\mathbf{v}_1 = \mathbf{v}_0$ at time $t_1 = 0$, and its final velocity be $\mathbf{v}_2 = \mathbf{v}(t)$ at some arbitrary time t. Then from equation (8), we have

$$\mathbf{v}(t) - \mathbf{v}_0 = \mathbf{a}\int_0^t \mathrm{d}t$$

$$\implies \mathbf{v}(t) - \mathbf{v}_0 = \mathbf{a}(t-0)$$

$$\implies \boxed{\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}t} \tag{9}$$

which is the *velocity-time relation*, or the *first kinematic relation*.

1.1.2 Position-time relation

Consider the motion of a particle moving with a constant acceleration $\mathbf{a}$. Let its initial position be $\mathbf{r}_1 = \mathbf{r}_0$ at time $t_1 = 0$, and its final position be $\mathbf{r}_2 = \mathbf{r}(t)$ at some arbitrary time t. Then from

equation (5), we have

$$\begin{aligned}
\mathbf{r}(t)-\mathbf{r}_0 &= \int_0^t \mathbf{v}\,\mathrm{d}t \\
&= \int_0^t (\mathbf{v}_0+\mathbf{a}t)\,\mathrm{d}t \quad \text{[using eqn. (9)]} \\
&= \mathbf{v}_0\int_0^t \mathrm{d}t + \mathbf{a}\int_0^t t\,\mathrm{d}t \\
&= \mathbf{v}_0(t-0)+\mathbf{a}\,\frac{t^2}{2}\bigg|_0^t \\
\implies &\boxed{\mathbf{r}(t)=\mathbf{r}_0+\mathbf{v}_0 t+\frac{1}{2}\mathbf{a}t^2} \qquad (10)
\end{aligned}$$

which is the *position-time relation*, or the *second kinematic relation*.

1.1.3 Position-velocity relation

We know that

$$\mathbf{a}=\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$$

Taking the dot product of the above with an infinitessimal displacement d$\mathbf{r}$, we have

$$\begin{aligned}
\mathbf{a}\cdot\mathrm{d}\mathbf{r} &= \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathrm{d}\mathbf{r} \\
\implies \mathbf{a}\cdot\mathrm{d}\mathbf{r} &= \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\cdot\mathrm{d}\mathbf{v} \\
\implies \mathbf{a}\cdot\mathrm{d}\mathbf{r} &= \mathbf{v}\cdot\mathrm{d}\mathbf{v} \qquad (11)
\end{aligned}$$

Now, using the product rule for differentiation, we have

$$\begin{aligned}
\mathrm{d}(\mathbf{v}\cdot\mathbf{v}) &= \mathrm{d}\mathbf{v}\cdot\mathbf{v}+\mathbf{v}\cdot\mathrm{d}\mathbf{v} \\
\implies 2\mathbf{v}\cdot\mathrm{d}\mathbf{v} &= \mathrm{d}(\mathbf{v}\cdot\mathbf{v}) \\
\implies \mathbf{v}\cdot\mathrm{d}\mathbf{v} &= \frac{1}{2}\mathrm{d}(\mathbf{v}\cdot\mathbf{v})
\end{aligned}$$

Then equation (11) becomes

$$\mathbf{a}\cdot\mathrm{d}\mathbf{r}=\frac{1}{2}\mathrm{d}(\mathbf{v}\cdot\mathbf{v}) \qquad (12)$$

Suppose we have a particle moving under constant acceleration $\mathbf{a}$. If at its initial and final positions $\mathbf{r}_0$ and $\mathbf{r}$, it has initial and final velocities $\mathbf{v}_0$ and $\mathbf{v}$, then we can integrate equation (eqn:adr-

vdv-02) as

$$\begin{aligned}
\mathbf{a}\cdot\int_{\mathbf{r}_0}^{\mathbf{r}}\mathrm{d}\mathbf{r} &= \frac{1}{2}\int_{\mathbf{v}_0}^{\mathbf{v}}\mathrm{d}(\mathbf{v}\cdot\mathbf{v}) \\
\implies \mathbf{a}\cdot\mathbf{r}\big|_{\mathbf{r}_0}^{\mathbf{r}} &= \frac{1}{2}\,\mathbf{v}\cdot\mathbf{v}\big|_{\mathbf{v}_0}^{\mathbf{v}} \\
\implies \mathbf{a}\cdot(\mathbf{r}-\mathbf{r}_0) &= \frac{1}{2}(\mathbf{v}\cdot\mathbf{v}-\mathbf{v}_0\cdot\mathbf{v}_0) \\
&= \frac{1}{2}\left(v^2-v_0^2\right) \\
\implies \boxed{v^2 = v_0^2+2\mathbf{a}\cdot\left(v^2-v_0^2\right)} &
\end{aligned} \tag{13}$$

which is the *position-velocity relation*, or the *third kinematic relation*.

1.2 Newton's laws of motion

1.2.1 Newton's first law of motion

A particle tends to maintain its state of rest or uniform motion unless it is acted upon by an external agency.

- The importance of Newton's first law of motion lies in that it defines force as:

  Force is any external agency that brings about a change in the state of rest or uniform motion of a particle.

1.2.2 Newton's second law of motion

The rate of change of momentum of a particle is directly proportional to the force exerted on it, and takes place in the direction of the force.

$$\mathbf{F}\propto\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$$

This implies that we have

$$\mathbf{F}=k\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$$

where k is the proportionality constant.

Now, for a particle of mass m and moving with a velocity $\mathbf{v}$, its momentum is defined as $\mathbf{p}=m\mathbf{v}$. Hence, Newton's second law becomes

$$\begin{aligned}
\mathbf{F} &= k\frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}) \\
&= km\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \quad \text{[assuming } m \text{ to be constant]} \\
\mathbf{F} &= km\mathbf{a}
\end{aligned}$$

The constant k is set to 1 by our choice of the units of force. We define 1 newton to be that amount of force that brings about an acceleration of 1 m/s^2 on a particle of mass 1 kg. Then from the above, we have

$$\begin{aligned}(1\text{ N}) &= k \times \left(1\text{ kg}\right) \times \left(1\text{ m/s}^2\right) \\ \implies k &= 1\end{aligned}$$

Therefore, for a particle *whose mass remains unchanged*, Newton's second law of motion can be written as

$$\boxed{\mathbf{F} = m\mathbf{a}}$$

- The importance of Newton's second law of motion lies in that it measures force as:

  If a force of magnitude F brings about an acceleration a on a particle of mass m, then the force is measured to be $F = ma$.

1.2.3 Newton's third law of motion

For every action, there is an equal and opposite reaction, both of which act on different bodies.

$$\mathbf{F}_{12} = -\mathbf{F}_{21}$$

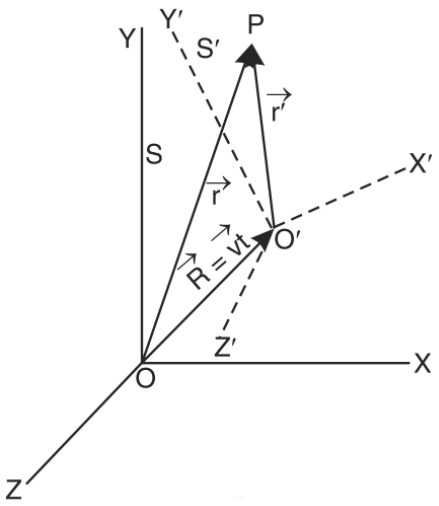
- The importance of Newton's second law of motion lies in that it gives the origin of force as:

  If a particle experiences a force of magnitude F, it is because it exerts another force of the same magnitude F but in the opposite direction on some other particle.

2 Inertial frames of reference

A reference frame (e.g. a coordinate system) in which Newton's first law of motion holds good, is known as a *Galilean* or *inertial frame of reference*. Hence, in an inertial frame a body continues in its state of rest or of uniform motion as long as no external force acts on it.

All frames of reference moving with a constant velocity with respect to an inertial frame are also inertial frames of reference.



Consider an inertial frame of reference S and another reference frame S', such that their orientations in space remain fixed and are thus free to have only translatory motion with respect to one another. Suppose further that the origins O and O' at the two systems initially coincident.

As shown in the figure the frame S' has moved into its current position in time t with a constant velocity $\mathbf{v}$ relative to S. Then we have

$$\bar{OO'} = \mathbf{R} = \mathbf{v}t$$

The position vectors $\mathbf{r}$ and $\mathbf{r}'$ of a particle P, with respect to the two frames of reference are then related as

$$\begin{aligned} \mathbf{r} &= \mathbf{R} + \mathbf{r}' \\ \implies \mathbf{r}' &= \mathbf{r} - \mathbf{R} \\ \implies \mathbf{r}' &= \mathbf{r} - \mathbf{v}t \end{aligned}$$

Differentiating the above, we get

$$\frac{\mathrm{d}\mathbf{r}'}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} - \mathbf{v}$$

Differentiating once again, we get

$$\begin{aligned} \frac{\mathrm{d}^2\mathbf{r}'}{\mathrm{d}t^2} &= \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} \\ \implies \mathbf{a}' &= \mathbf{a} \end{aligned}$$

Hence, the acceleration of the particle is the same in the two frames of reference. This indicates that if the particle is at rest or uniform motion in the frame S, then it would also appear to be at rest or uniform motion in the other frame S'.

Therefore, all frames of reference, moving with a constant velocity with respect to an inertial frame, are also inertial frames of reference.

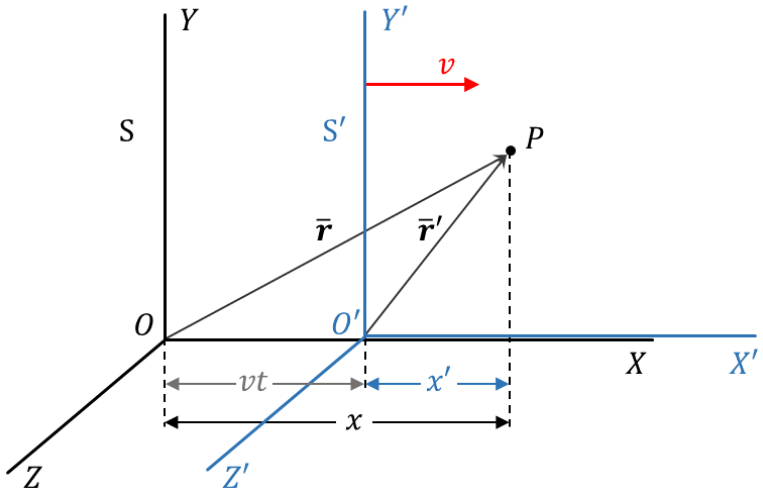
Examples of inertial frames of reference:

1. An object in deep space, away from any gravitational influence, can be considered to be in an inertial frame of reference.
2. An object falling freely under the influence of gravity can be considered to be in an inertial frame of reference.

3 Galilean transformation

Suppose an event occurs at some point in space, and it is being observed from two different frames of reference. The we would have two different sets of coordinates which correspond to the same event, as observed from both frames of reference. The set of equations that relate these two sets of coordinates are known as *transformation equations*, because they enable us to *transform* a set of coordinates in one system to another set of coordinates in a different system.

The set of equations used to convert the space and time coordinates of a particle from one inertial frame of reference to another moving at a constant relative velocity is known as a *Galilean transformation*.



Let S and S' be two inertial frames of reference whose axes and origins coincide and clocks are synchronized at $t = 0$. Suppose the frame S' is moving along the positive X direction with a constant relative speed v with respect to the frame S. Then, at some arbitrary time t, the separation between the frames is $v\,t$.

Now, suppose some event occurs at the position P at time t. Then $\mathbf{r}$ and $\mathbf{r}'$ are the position vectors of the same event with respect to the frames S and S' respectively. If x and x' are the X-coordinates of P with respect to S and S' respectively, then they are related as

$$x = x' + v\,t$$

Also, because there is no relative motion between the frames along the Y and Z directions, these coordinates with respect to the two frames would be

$$y = y' \quad \text{and} \quad z = z'$$

Finally, because the clocks of both frames are synchronized, they measure identical times, and so we would have

$$t = t'$$

Therefore, the four *Galilean transformation equations* to transform from frame S to frame S' are

$$\boxed{\begin{aligned} x' &= x - v\,t \\ y' &= y \\ z' &= z \\ t' &= t \end{aligned}}$$

In vector form, the above transformation equations may be combined as

$$\boxed{\mathbf{r}' = \mathbf{r} - \mathbf{v}t}$$

The four *inverse Galilean transformation equations* to transform from frame S' to frame S are

$$\boxed{\begin{aligned} x &= x' + v\,t' \\ y &= y' \\ z &= z' \\ t &= t' \end{aligned}}$$

In vector form, the above inverse transformation equations may be combined as

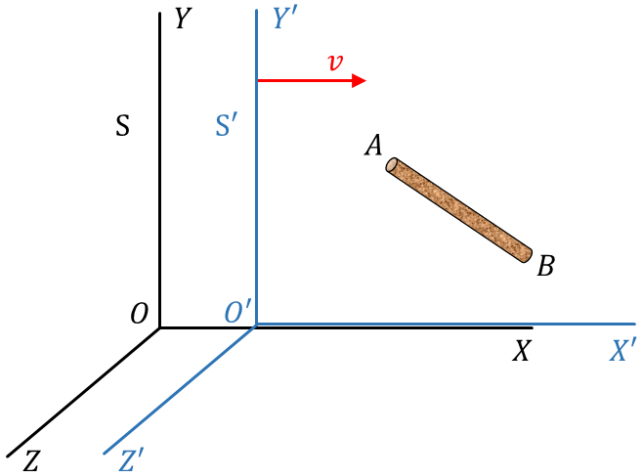
$$\boxed{\mathbf{r} = \mathbf{r}' + \mathbf{v}t'}$$

3.1 Galilean invariance

When a physical quantity has the same value in two frames of reference, we say that the quantity is *invariant* to a transformation between the frames.

3.1.1 Transformation of length

Consider a rod with ends A and B, which is being observed with respect to two inertial frames S and S'. The frame S' is moving with a constant speed v relative to the frame S along the positive X direction.



In the frame S, the coordinates of the ends of the rod are:

$$A : (x_A, y_A, z_A)$$
$$B : (x_B, y_B, z_B)$$

Likewise, in the frame S', the coordinates of the ends of the rod are:

$$A : (x'_A, y'_A, z'_A)$$
$$B : (x'_B, y'_B, z'_B)$$

The length of the rod is the distance between A and B. Hence in the frame S, we have

$$L = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2 + (z_A - z_B)^2} \tag{1}$$

Similarly, in the frame S', we have

$$L' = \sqrt{(x'_A - x'_B)^2 + (y'_A - y'_B)^2 + (z'_A - z'_B)^2} \tag{2}$$

Now, from the Galilean transformation of the coordinates of A and B, we have

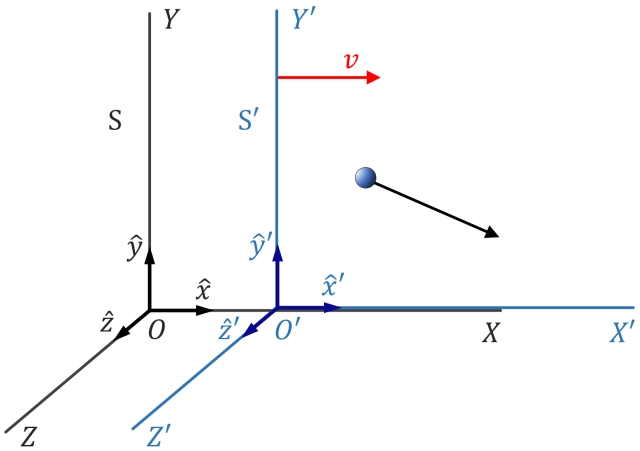
$$x'_A = x_A - v\,t,\ y'_A = y_A,\ z'_A = z_A$$
$$x'_B = x_B - v\,t,\ y'_B = y_B,\ z'_B = z_B$$

Using the above in equation (2), we get

$$\begin{aligned} L' &= \sqrt{[(x_A - v\,t) - (x_B - v\,t)]^2 + (y_A - y_B)^2 + (z_A - z_B)^2} \\ &= \sqrt{[x_A - \cancel{v\,t} - x_B - \cancel{v\,t}]^2 + (y_A - y_B)^2 + (z_A - z_B)^2} \\ &= \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2 + (z_A - z_B)^2} \\ \implies & \boxed{L' = L} \quad \text{[using eqn. (1)]} \end{aligned}$$

The above equation shows us that *length or distance between two points remains invariant to a Galilean transformation.*

3.1.2 Transformation of velocity



Consider a particle in motion, which is being observed from two inertial frames of reference S and S'. Frame S' is moving with a constant speed v along the positive X direction relative to the frame S. This implies that the unit vector basis of the two frames always remain correspondingly parallel, and so we have

$$\hat{\mathbf{x}}' = \hat{\mathbf{x}},\ \hat{\mathbf{y}}' = \hat{\mathbf{y}},\ \hat{\mathbf{z}}' = \hat{\mathbf{z}} \tag{1}$$

Also, the velocity of the frame S' with respect to S is

$$\mathbf{v} = \hat{\mathbf{x}}v \tag{2}$$

From the Galilean transformation for the time coordinate, we have

$$\begin{aligned} t' &= t \\ \implies \frac{\mathrm{d}t'}{\mathrm{d}t} &= 1 \\ \text{or,} \quad \frac{\mathrm{d}t}{\mathrm{d}t'} &= 1 \end{aligned}$$

Now, the Galilean transformation for the X coordinate is

$$\begin{aligned} x' &= x - v\,t \\ \implies \frac{\mathrm{d}x'}{\mathrm{d}t'} &= \frac{\mathrm{d}}{\mathrm{d}t'}(x - v\,t) \\ &= \frac{\mathrm{d}x}{\mathrm{d}t'} - \frac{\mathrm{d}}{\mathrm{d}t'}(v\,t) \\ &= \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}t'} - v\frac{\mathrm{d}t}{\mathrm{d}t'} \\ \implies \frac{\mathrm{d}x'}{\mathrm{d}t'} &= \frac{\mathrm{d}x}{\mathrm{d}t} - v \end{aligned}$$

Likewise, we have for the Y and Z coordinates

$$\begin{aligned} y' &= y \quad \text{and} \quad z' = z \\ \implies \frac{\mathrm{d}y'}{\mathrm{d}t'} &= \frac{\mathrm{d}y}{\mathrm{d}t'} \quad \text{and} \quad \frac{\mathrm{d}z'}{\mathrm{d}t'} = \frac{\mathrm{d}z}{\mathrm{d}t'} \\ \implies \frac{\mathrm{d}y'}{\mathrm{d}t'} &= \frac{\mathrm{d}y}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}t'} \quad \text{and} \quad \frac{\mathrm{d}z'}{\mathrm{d}t'} = \frac{\mathrm{d}z}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}t'} \\ \implies \frac{\mathrm{d}y'}{\mathrm{d}t'} &= \frac{\mathrm{d}y}{\mathrm{d}t} \quad \text{and} \quad \frac{\mathrm{d}z'}{\mathrm{d}t'} = \frac{\mathrm{d}z}{\mathrm{d}t} \end{aligned}$$

The velocity of the particle, as observed from the frame S, is given as

$$\mathbf{u} = \hat{\mathbf{x}}\frac{\mathrm{d}x}{\mathrm{d}t} + \hat{\mathbf{y}}\frac{\mathrm{d}y}{\mathrm{d}t} + \hat{\mathbf{z}}\frac{\mathrm{d}z}{\mathrm{d}t} \tag{3}$$

And its velocity, as observed from the frame S', is given as

$$\begin{aligned} \mathbf{u}' &= \hat{\mathbf{x}}'\frac{\mathrm{d}x'}{\mathrm{d}t'} + \hat{\mathbf{y}}'\frac{\mathrm{d}y'}{\mathrm{d}t'} + \hat{\mathbf{z}}'\frac{\mathrm{d}z'}{\mathrm{d}t'} \\ &= \hat{\mathbf{x}}'\left(\frac{\mathrm{d}x}{\mathrm{d}t} - v\right) + \hat{\mathbf{y}}'\frac{\mathrm{d}y}{\mathrm{d}t} + \hat{\mathbf{z}}'\frac{\mathrm{d}z}{\mathrm{d}t} \\ &= \hat{\mathbf{x}}\left(\frac{\mathrm{d}x}{\mathrm{d}t} - v\right) + \hat{\mathbf{y}}\frac{\mathrm{d}y}{\mathrm{d}t} + \hat{\mathbf{z}}\frac{\mathrm{d}z}{\mathrm{d}t} \quad \text{[using eqn. (1)]} \\ &= \left(\hat{\mathbf{x}}\frac{\mathrm{d}x}{\mathrm{d}t} + \hat{\mathbf{y}}\frac{\mathrm{d}y}{\mathrm{d}t} + \hat{\mathbf{z}}\frac{\mathrm{d}z}{\mathrm{d}t}\right) - (\hat{\mathbf{x}}v) \\ \implies &\boxed{\mathbf{u}' = \mathbf{u} - \mathbf{v}} \quad \text{[using eqns. (2) and (3)]} \end{aligned} \tag{4}$$

Equation (4) shows us that the *velocity of a particle is not invariant to a Galilean transformation.* It is the transformation of velocity across the inertial frames.

The inverse velocity transformation is

$$\boxed{\mathbf{u} = \mathbf{u}' + \mathbf{v}} \tag{5}$$

Equations (4) and (5) are referred to as the Galilean law of addition of velocities.

3.1.3 Transformation of acceleration

The acceleration of a particle measured with respect to the frame S is

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t}$$

Likewise, the acceleration of the same particle measured with respect to the frame S' is

$$\mathbf{a}' = \frac{\mathrm{d}\mathbf{u}'}{\mathrm{d}t'} = \frac{\mathrm{d}\mathbf{u}'}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}t'} = \frac{\mathrm{d}\mathbf{u}'}{\mathrm{d}t}$$

Using the Galilean law of addition of velocities above, we get

$$\begin{aligned}\mathbf{a}' &= \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{u}-\mathbf{v}) \\ &= \frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} - \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\end{aligned}$$

Because the velocity $\mathbf{v}$ of the frame S' with respect to S remains constant, we have $\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = 0$. Then the above becomes

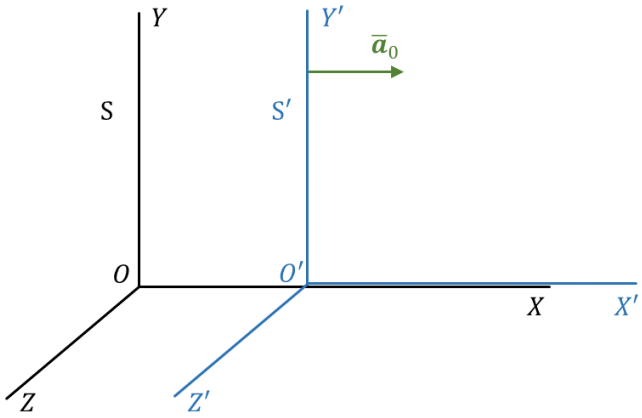
$$\mathbf{a}' = \frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t}$$
$$\boxed{\mathbf{a}' = \mathbf{a}}$$

The above equation shows us that *the acceleration of a particle remains invariant to a Galilean transformation.*

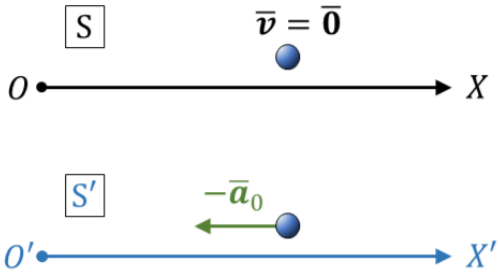
4 Non-inertial frames of reference

A reference frame (e.g. a coordinate system) in which Newton's first law of motion does not hold good, is known as a *non-inertial frame of reference*.

All frames of reference having an accelerated motion with respect to an inertial frame are non-inertial frames of reference.



In the above figure, S is an inertial frame. Another frame S' is moving with an acceleration $\mathbf{a}_0$ with respect to frame S. Here the frame S' is a non-inertial frame of reference.

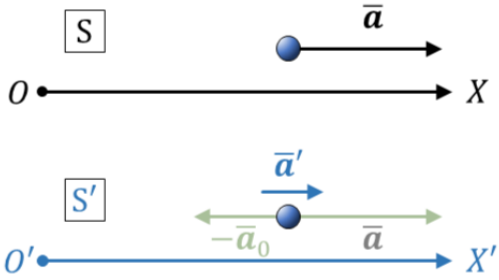


If a particle of mass m is at rest with respect to the frame S, then it would appear to have an acceleration $-\mathbf{a}_0$ with respect to the frame S'.

From Newton's second law of motion, with respect to S', it appears as if some pseudo-force

$$\mathbf{F}_p = -m\mathbf{a}_0 \tag{1}$$

is acting on the particle.



Suppose now that the particle has an acceleration $\mathbf{a}$ with respect to the frame S. Then the force acting on it in S is

$$\mathbf{F} = m\mathbf{a} \tag{2}$$

The acceleration of the particle with respect to the non-inertial frame S' would be

$$\mathbf{a}' = \mathbf{a} - \mathbf{a}_0$$

Then from Newton's second law of motion, the force acting on the particle with respect to S' would be

$$\begin{aligned} \mathbf{F}' &= m\mathbf{a}' \\ &= m(\mathbf{a} - \mathbf{a}_0) \\ &= m\mathbf{a} - m\mathbf{a}_0 \\ \implies &\boxed{\mathbf{F} = \mathbf{F} + \mathbf{F}_p} \quad \text{[using eqns. (1) and (2)]} \end{aligned} \tag{3}$$

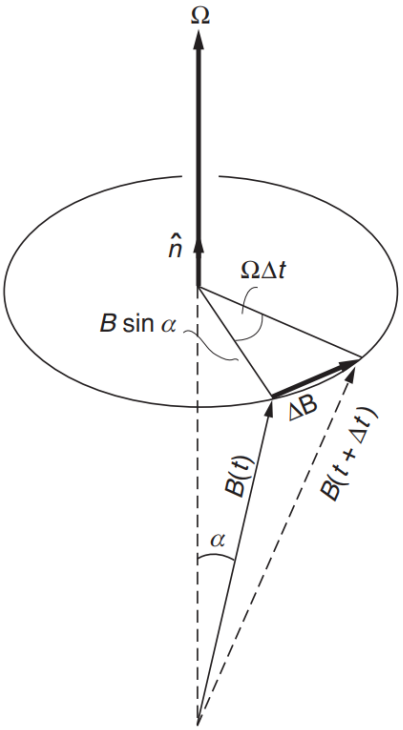
Equations (1) and (3) show that, regardless of whether a force is acting on a particle in the inertial frame, a pseudo-force of $\mathbf{F}_p = -m\mathbf{a}_0$ always seems to be acting on the particle in the non-inertial frame of reference.

Such pseudo-forces, which arise due to the accelerated motion of a non-inertial frame of reference, are known as *fictitious forces*.

4.1 Rotating coordinate systems

Rate of change of a rotating vector

Consider an arbitrary vector $\mathbf{B}$ which is precessing at an angular frequency of Ω about an arbitrary axis lying in the $\hat{\mathbf{n}}$ direction as shown in the figure.



Then the angular velocity vector is $\boldsymbol{\Omega} = \Omega\hat{\mathbf{n}}$. Let α be the angle between the vectors $\mathbf{B}$ and $\boldsymbol{\Omega}$. At times t and $t + \Delta t$, we have the vectors $\mathbf{B}(t)$ and $\mathbf{B}(t + \Delta t)$ respectively. The tip of $\mathbf{B}$ sweeps out a circular path with radius $B\sin\alpha$ and the plane of the circle is perpendicular to $\hat{\mathbf{n}}$.

During time Δt the tip of $\mathbf{B}$ swings through an angle of $\Omega\Delta t$. Consequently, the tip of $\mathbf{B}$ moves by an amount ΔB, where

$$\Delta B \approx (B\sin\alpha)\Omega\Delta t$$

Then the rate of change of $\mathbf{B}$ has the magnitude

$$\begin{aligned} \frac{\mathrm{d}B}{\mathrm{d}t} &= \lim_{\Delta t \to 0} \frac{\Delta B}{\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{(B\sin\alpha)\Omega\Delta t}{\Delta t} \\ \implies \frac{\mathrm{d}B}{\mathrm{d}t} &= \Omega B\sin\alpha \end{aligned}$$

From the figure, we find that $\Delta\mathbf{B}$ is perpendicular to both $\mathbf{B}$ and $\hat{\mathbf{n}}$. Hence $\dfrac{\mathrm{d}\mathbf{B}}{\mathrm{d}t}$ is also perpendicular to both $\mathbf{B}$ and $\hat{\mathbf{n}}$. We also find that $\Omega B\sin\alpha = |\boldsymbol{\Omega}\times\mathbf{B}|$. Therefore,

$$\frac{\mathrm{d}\mathbf{B}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\mathbf{B} \tag{1}$$

Equation (1) is true for any vector $\mathbf{B}$ that undergoes pure rotation around an axis $\hat{\mathbf{n}}$ at a rate Ω.

Time derivative of a vector in a rotating frame

We consider a frame of reference S' that is rotating with respect to an inertial frame S. Let the base vectors be $(\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}})$ in the inertial frame and $(\hat{\mathbf{x}}',\hat{\mathbf{y}}',\hat{\mathbf{z}}')$. Then some arbitrary vector $\mathbf{C}$ can be expressed in either frames as

$$\mathbf{C} = \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z = \hat{\mathbf{x}}'C'_x + \hat{\mathbf{y}}'C'_y + \hat{\mathbf{z}}'C'_z \tag{2}$$

The time derivative of $\mathbf{C}$ in the inertial frame is

$$\left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{in}} = \hat{\mathbf{x}}\frac{\mathrm{d}C_x}{\mathrm{d}t} + \hat{\mathbf{y}}\frac{\mathrm{d}C_y}{\mathrm{d}t} + \hat{\mathbf{z}}\frac{\mathrm{d}C_z}{\mathrm{d}t} \tag{3}$$

And the time derivative of $\mathbf{C}$ in the rotating frame is

$$\left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{rot}} = \hat{\mathbf{x}}'\frac{\mathrm{d}C'_x}{\mathrm{d}t} + \hat{\mathbf{y}}'\frac{\mathrm{d}C'_y}{\mathrm{d}t} + \hat{\mathbf{z}}'\frac{\mathrm{d}C'_z}{\mathrm{d}t} \tag{4}$$

Now the base vectors $(\hat{\mathbf{x}}',\hat{\mathbf{y}}',\hat{\mathbf{z}}')$ in the rotating frame all rotate with the same angular velocity $\boldsymbol{\Omega}$. Hence, in equation (1), replacing $\mathbf{B}$ with $\hat{\mathbf{x}}'$, $\hat{\mathbf{y}}'$ and $\hat{\mathbf{z}}'$, we have

$$\frac{\mathrm{d}\hat{\mathbf{x}}'}{\mathrm{d}t} = \boldsymbol{\Omega}\times\hat{\mathbf{x}}', \quad \frac{\mathrm{d}\hat{\mathbf{y}}'}{\mathrm{d}t} = \boldsymbol{\Omega}\times\hat{\mathbf{y}}', \quad \frac{\mathrm{d}\hat{\mathbf{z}}'}{\mathrm{d}t} = \boldsymbol{\Omega}\times\hat{\mathbf{z}}' \tag{5}$$

Taking the time derivative of equation (2) gives us

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t}\left(\hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\hat{\mathbf{x}}'C'_x + \hat{\mathbf{y}}'C'_y + \hat{\mathbf{z}}'C'_z\right)\\
\implies &\hat{\mathbf{x}}\frac{\mathrm{d}C_x}{\mathrm{d}t} + \hat{\mathbf{y}}\frac{\mathrm{d}C_y}{\mathrm{d}t} + \hat{\mathbf{z}}\frac{\mathrm{d}C_z}{\mathrm{d}t} = \left(\hat{\mathbf{x}}'\frac{\mathrm{d}C'_x}{\mathrm{d}t} + C'_x\frac{\mathrm{d}\hat{\mathbf{x}}'}{\mathrm{d}t}\right) + \left(\hat{\mathbf{y}}'\frac{\mathrm{d}C'_y}{\mathrm{d}t} + C'_y\frac{\mathrm{d}\hat{\mathbf{y}}'}{\mathrm{d}t}\right) + \left(\hat{\mathbf{z}}'\frac{\mathrm{d}C'_z}{\mathrm{d}t} + C'_z\frac{\mathrm{d}\hat{\mathbf{z}}'}{\mathrm{d}t}\right)\\
\implies &\hat{\mathbf{x}}\frac{\mathrm{d}C_x}{\mathrm{d}t} + \hat{\mathbf{y}}\frac{\mathrm{d}C_y}{\mathrm{d}t} + \hat{\mathbf{z}}\frac{\mathrm{d}C_z}{\mathrm{d}t} = \left(\hat{\mathbf{x}}'\frac{\mathrm{d}C'_x}{\mathrm{d}t} + \hat{\mathbf{y}}'\frac{\mathrm{d}C'_y}{\mathrm{d}t} + \hat{\mathbf{z}}'\frac{\mathrm{d}C'_z}{\mathrm{d}t}\right) + \left(C'_x\frac{\mathrm{d}\hat{\mathbf{x}}'}{\mathrm{d}t} + C'_y\frac{\mathrm{d}\hat{\mathbf{y}}'}{\mathrm{d}t} + C'_z\frac{\mathrm{d}\hat{\mathbf{z}}'}{\mathrm{d}t}\right)\\
&\implies \left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{in}} = \left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{rot}} + \left(C'_x\boldsymbol{\Omega}\times\hat{\mathbf{x}}' + C'_y\boldsymbol{\Omega}\times\hat{\mathbf{y}}' + C'_z\boldsymbol{\Omega}\times\hat{\mathbf{z}}'\right) \quad \text{[Using eqns. (3), (4), (5)]}\\
&\implies \left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{in}} = \left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{rot}} + \boldsymbol{\Omega}\times\left(\hat{\mathbf{x}}'C'_x + \hat{\mathbf{y}}'C'_y + \hat{\mathbf{z}}'C'_z\right)\\
&\implies \left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{in}} = \left(\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t}\right)_{\text{rot}} + \boldsymbol{\Omega}\times\mathbf{C} \quad \text{[Using eqn. (2)]}
\end{aligned} \tag{6}$$

4.1.1 Velocity and acceleration in a rotating frame

In equation (6), if we set **C** as the position vector of the particle, i.e. $\mathbf{C}=\mathbf{r}$, then we have

$$\left(\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\right)_{\text{in}}=\left(\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\right)_{\text{rot}}+\boldsymbol{\Omega}\times\mathbf{r}$$

$$\implies \mathbf{v}_{\text{in}}=\mathbf{v}_{\text{rot}}+\boldsymbol{\Omega}\times\mathbf{r} \tag{7}$$

$$\implies \boxed{\mathbf{v}_{\text{rot}}=\mathbf{v}_{\text{in}}-\boldsymbol{\Omega}\times\mathbf{r}} \tag{8}$$

where $\mathbf{v}_{\text{in}}$, $\mathbf{v}_{\text{rot}}$ are the velocities of the particle measured with respect to S and S' respectively.

Again, in equation (6), if we set **C** as the velocity of the particle in the inertial frame, i.e. $\mathbf{C}=\mathbf{v}_{\text{in}}$, then we have

$$\begin{aligned}\left(\frac{\mathrm{d}\mathbf{v}_{\text{in}}}{\mathrm{d}t}\right)_{\text{in}}&=\left(\frac{\mathrm{d}\mathbf{v}_{\text{in}}}{\mathrm{d}t}\right)_{\text{rot}}+\boldsymbol{\Omega}\times\mathbf{v}_{\text{in}}\\&=\left[\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{v}_{\text{rot}}+\boldsymbol{\Omega}\times\mathbf{r})\right]_{\text{rot}}+\boldsymbol{\Omega}\times[\mathbf{v}_{\text{rot}}+\boldsymbol{\Omega}\times\mathbf{r}]\quad[\text{Using eqn. (7)}]\\&=\left(\frac{\mathrm{d}\mathbf{v}_{\text{rot}}}{\mathrm{d}t}\right)_{\text{rot}}+\boldsymbol{\Omega}\times\underbrace{\left(\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\right)_{\text{rot}}}_{=\mathbf{v}_{\text{rot}}}+\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}+\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})\end{aligned}$$

$$\implies \underbrace{\left(\frac{\mathrm{d}\mathbf{v}_{\text{in}}}{\mathrm{d}t}\right)_{\text{in}}}_{=\mathbf{a}_{\text{in}}}=\underbrace{\left(\frac{\mathrm{d}\mathbf{v}_{\text{rot}}}{\mathrm{d}t}\right)_{\text{rot}}}_{=\mathbf{a}_{\text{rot}}}+\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}+\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}+\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})$$

$$\implies \mathbf{a}_{\text{in}}=\mathbf{a}_{\text{rot}}+2\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}+\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})$$

$$\implies \boxed{\mathbf{a}_{\text{rot}}=\mathbf{a}_{\text{in}}-2\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}-\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})} \tag{9}$$

Equations (8) and (9) respectively give us the velocity and acceleration of the particle as observed from the rotating frame of reference.

4.1.2 Fictitious forces in a rotating frame

With respect to a rotating frame of reference, Newton's second law of motion gives us the net force acting on a particle of mass m to be

$$\mathbf{F}_{\text{rot}}=m\mathbf{a}_{\text{rot}} \tag{10}$$

Inserting equation (9) into equation (10) we obtain

$$\begin{aligned}\mathbf{F}_{\text{rot}}&=m[\mathbf{a}_{\text{in}}-2\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}-\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})]\\&=m\mathbf{a}_{\text{in}}-2m\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})\\\implies \mathbf{F}_{\text{rot}}&=\mathbf{F}-2m\boldsymbol{\Omega}\times\mathbf{v}_{\text{rot}}-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})\end{aligned} \tag{11}$$

On the RHS of equation (11), we refer to all the terms as

- *Actual force:* $\mathbf{F}=m\mathbf{a}_{\text{in}}$
  This is the force on the particle as observed from the inertial frame of reference.

- *Centrifugal force:* $\mathbf{F}_{\text{centrifugal}} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$
  This fictitious force is perpendicular to the axis of rotation and is directed radially outward.
- *Coriolis force:* $\mathbf{F}_{\text{Coriolis}} = -2m\boldsymbol{\Omega} \times \mathbf{v}_{\text{rot}}$
  This fictitious force is an apparent sideways force that deflects a moving object perpendicularly to both the direction of rotation and the object's own motion.

Therefore, we can write the force on the particle in a rotating frame as

$$\mathbf{F}_{\text{rot}} = \mathbf{F} + \mathbf{F}_{\text{centrifugal}} + \mathbf{F}_{\text{Coriolis}}$$
$$\text{or,} \quad \mathbf{F}_{\text{rot}} = \mathbf{F}_{\text{fict}}$$

where the total fictitious force due to the rotating frame is

$$\mathbf{F}_{\text{fict}} = \mathbf{F}_{\text{centrifugal}} + \mathbf{F}_{\text{Coriolis}}$$